B25ET1005 OOP, Practical-4.

FAQ's.

1) What is the difference between a class and an object?

→ A class is a user defined blueprint or data type, whereas an object is a concrete run-time instance of that class.

2) Why are the real & imaginary data members kept inside the class?

→ To enforce data encapsulation & protect internal representations from direct external manipulation.

3) What does a constructor do?

→ It initializes an object into a valid state automatically at the time of its creation.

4) How is division of two complex numbers calculated?

→ By multiplying the numerator and denominator by the complex conjugate of the divisor to eliminate the imaginary term from the denominator.

5) What is a complex conjugate?

→ It is a complex number with the same real part but an inverted sign for its imaginary part (a - jb).

6) What are data members?
→ They are the variable declared inside a class that hold the state or data of an object.

7) What are member functions?
→ They are functions defined inside a class that operate on its data members.

8) What is encapsulation?
→ It is the bundling of data and the methods that operate on that data into a single unit while restricting direct access.

9) What is data hiding?
→ It is the mechanism of keeping data members private to prevent unauthorized external access.

10) What is abstraction?
→ It is the process/concept of exposing essential features of an entity while hiding its internal implementation details.